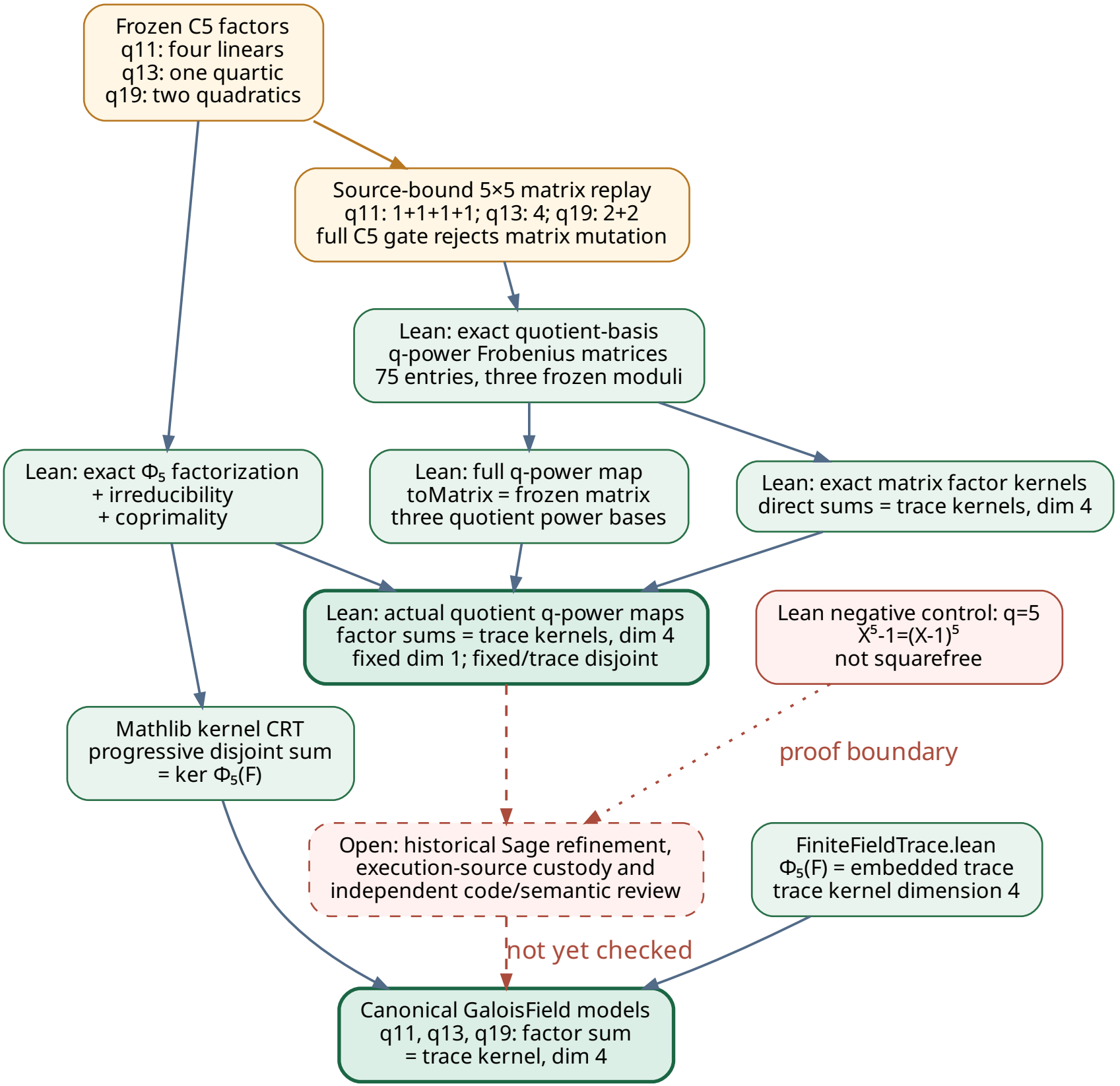
Frozen C5 factors
q11: four linears
q13: one quartic
q19: two quadratics
Source-bound 5×5 matrix replay
q11: 1+1+1+1; q13: 4; q19: 2+2
full C5 gate rejects matrix mutation
Lean: exact quotient-basis
q-power Frobenius matrices
75 entries, three frozen moduli
Lean: exact Φ5 factorization
+ irreducibility
+ coprimality
Lean: full q-power map
toMatrix = frozen matrix
three quotient power bases
Lean: exact matrix factor kernels
direct sums = trace kernels, dim 4
Lean: actual quotient q-power maps
factor sums = trace kernels, dim 4
fixed dim 1; fixed/trace disjoint
Lean negative control: q=5
X5-1=(X-1)5
not squarefree
proof boundary
Mathlib kernel CRT
progressive disjoint sum
= ker Φ5(F)
Open: historical Sage refinement,
execution-source custody and
independent code/semantic review
FiniteFieldTrace.lean
Φ5(F) = embedded trace
trace kernel dimension 4
not yet checked
Canonical GaloisField models
q11, q13, q19: factor sum
= trace kernel, dim 4